

SemEval 2026 Task 4: Narrative Similarity

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Abstract

This report details our Milestone 1 progress. Standard cross-encoder classification architectures (DeBERTa) overfit to surface-level lexical cues, yielding a sub-optimal 52% test accuracy on Track A. To address this domain shift, we transitioned to a metric-learning approach. By fine-tuning Bi-Encoders (MiniLM, BGE-Large) using Contrastive Learning (Triplet Loss and Multiple Negatives Ranking Loss), we forced the models to learn global narrative structures rather than pairwise shortcuts. This approach achieved 59% accuracy across both tracks, demonstrating that robust independent representations significantly outperform noisy pairwise interactions when dealing with biased training data.

1 Introduction

Narrative similarity requires understanding abstract themes and resolutions. It goes beyond simple word overlap. This task features two tracks. Track A involves a binary choice between two candidates. Track B requires creating standalone story embeddings.

We initially expected cross encoders to perform best. However, experiments showed they overfit to synthetic data shortcuts. The training data contains unnatural peaks and high word overlap. This report describes our diagnostic process. We show how we moved from basic classification to a contrastive framework. This change successfully separates narrative reasoning from keyword matching.

2 State-of-the-Art Methods

2.1 Track A: Relational Judgment

- **Hybrid Neuro Symbolic Cascades:** Models like CascadeMind [1] combine LLM semantic power with hard rule based logic. They use neural self consistency voting where the LLM evaluates the prompt multiple times. If votes result in a tie, a symbolic module calculates lexical overlap, event chains, and tension curves to make the final decision.
- **Multi Persona LLM Debating:** This approach [2] embraces ambiguity by using multiple distinct system personas. One LLM acts as a formal critic while another acts as a casual reader. They evaluate the same triple and vote, mimicking a diverse human jury to create highly robust predictions.

2.2 Track B: Narrative Representation

- **Story Grammar Segmentation:** Standard models treat texts as bags of concepts, but this method forces embeddings to respect the timeline. It segments the story into five chronological phases from setting to resolution. It generates a dense vector for each phase and compares aligned phases using cosine similarity.

- **Contrastive Learning on Reformulations:** Strong embeddings are built by training models on structural rewrites. An LLM rewrites a story changing the vocabulary but strictly preserving the theme and outcome. Contrastive learning pulls these embeddings tightly together so the encoder learns the plot rather than surface words.

3 Exploratory Data Analysis

Before training, we conducted an extensive Exploratory Data Analysis (EDA) comparing the 200 human-annotated Dev items against the 1,900 LLM-generated synthetic training triples. Our goal was to identify potential domain shifts that could cause models to overfit. We uncovered severe generative biases across lexical, structural, and latent dimensions.

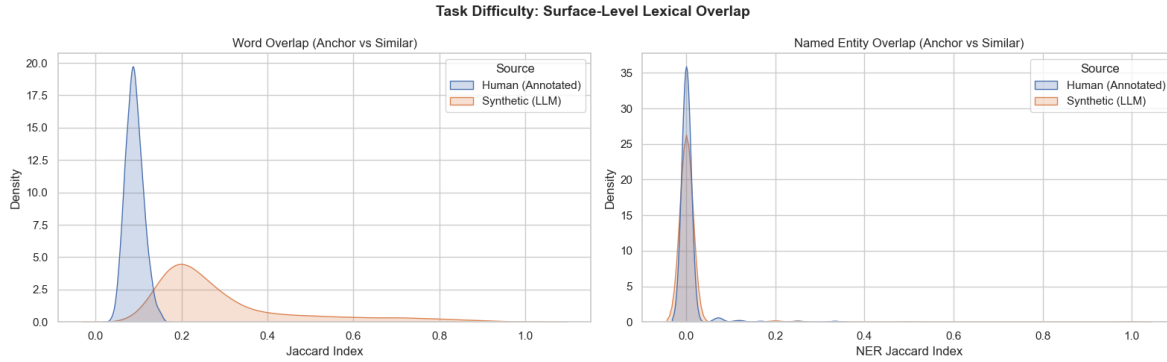


Figure 1: Density plot of Jaccard Word Overlap. The synthetic data (orange) provides highly overlapping lexical shortcuts not present in the human data (blue).

3.1 Generative Boilerplate and Phrasing

Furthermore, the synthetic data suffers from severe n-gram repetition. As shown in Figure 2, the LLM-generated anchors rely heavily on repetitive, formulaic transitions (e.g., “the film concludes with”, “delves deeper into the”). Conversely, the human-annotated anchors utilize natural, varied phrasing (e.g., “falls in love with”, “turns out to be”). A neural network trained on this synthetic data risks learning to trigger its attention weights on these specific boilerplate phrases rather than genuine plot markers.

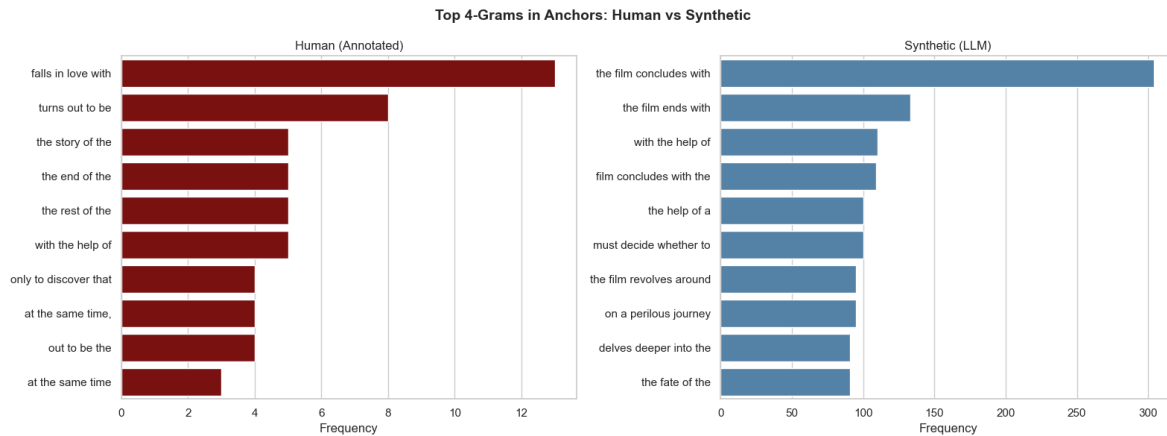


Figure 2: Top 4-Grams in Anchor texts. Synthetic text is dominated by repetitive LLM transitions compared to human variation.

3.2 Latent Space Domain Shift

To visualize the cumulative effect of these structural and lexical biases, we mapped the Anchor texts from both datasets into a shared vector space using Principal Component Analysis (PCA) on baseline embeddings (Figure 3).

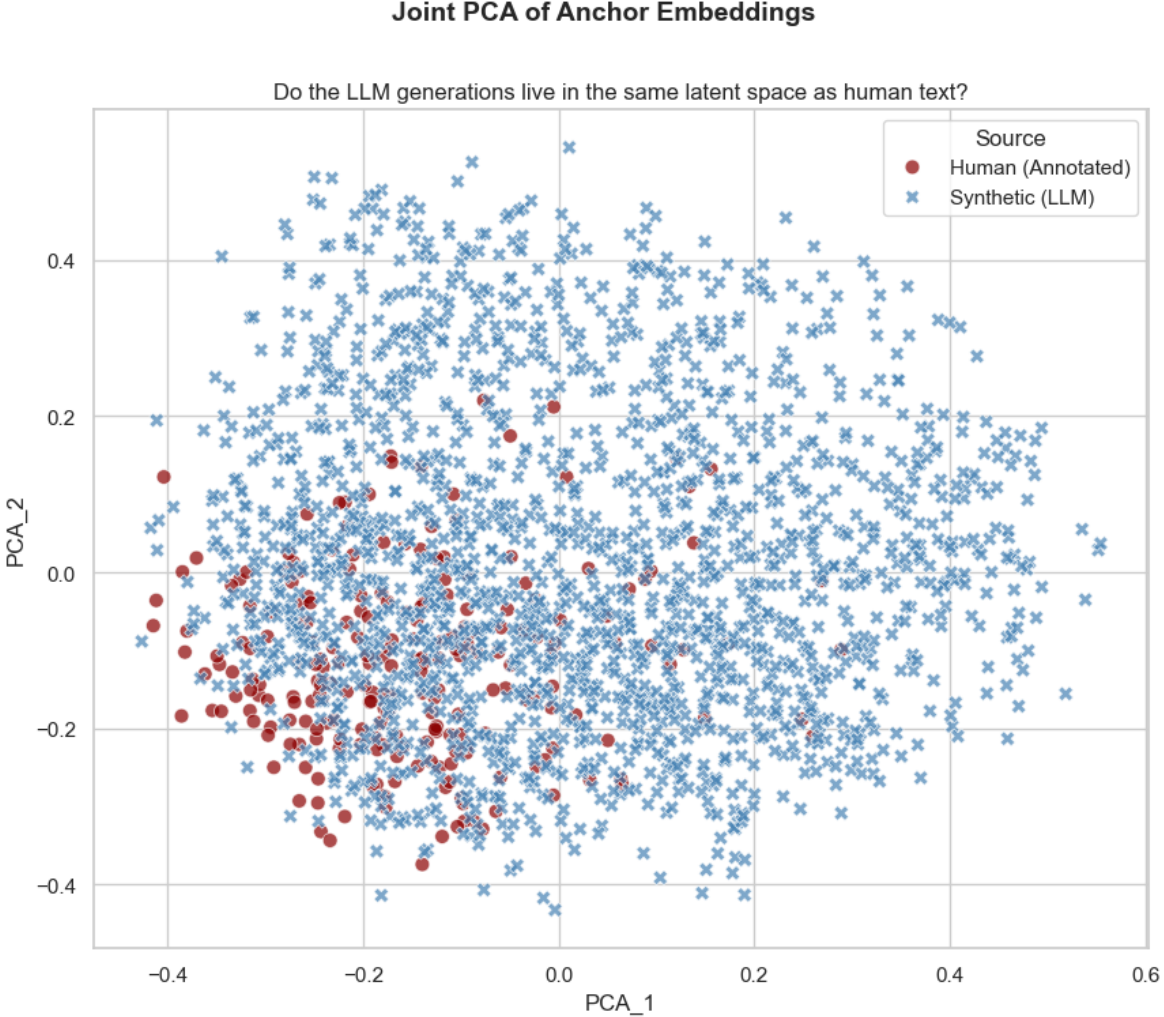


Figure 3: Joint PCA of Anchor Embeddings. The human texts (red) cluster tightly in a specific sub-space, proving a severe domain shift from the broad synthetic distribution (blue).

The scatter plot confirms a clear domain shift: the human-annotated texts (red) occupy a distinct, dense sub-region of the latent space, while the synthetic texts (blue) are broadly scattered. This ultimate finding confirmed our hypothesis: a Cross-Encoder trained exclusively on the synthetic manifold will struggle to generalize to the specific density and variance of human narrative logic.

4 Experiments and Results

To evaluate our hypothesis regarding generative bias and contrastive representation, we systematically tested a series of baselines, cross-encoders, and bi-encoders. Table 1 summarizes the performance across both tracks. Note that cross-encoders and generative LLMs are excluded from Track B, as the rules strictly require independent vector representations generated without pairwise context.

Model	Param	Track A (Acc)	Track B (Acc)
bge-large-en-v1.5	330M	0.680	0.652
all-MiniLM-L6-v2 (Tuned)	22M	0.662	0.592
Qwen-2.5-7B-instruct	7B	0.560	-
nli-deberta-v3-large	304M	0.520	-
uhhlt/story-emb	7B	0.520	-
all-MiniLM-L6-v2 (Zero-shot)	22M	0.500	-
deberta-v3-large	304M	0.490	-
Jaccard (Baseline)	-	0.560	-

Table 1: Performance comparison of baseline, cross-encoder, and contrastively fine-tuned models. Track B requires standalone vector representations, prohibiting cross-encoder and direct LLM evaluation.

4.1 Baseline and Cross-Encoder Failures

Our trivial baselines performed as expected for a complex reasoning task. The Jaccard lexical overlap baseline achieved 0.56 accuracy, while an off-the-shelf, non-fine-tuned all-MiniLM-L6-v2 achieved 0.50, representing random guessing.

When we fine-tuned standard cross-encoders using binary classification loss on the synthetic dataset, both deberta-v3-large (0.49) and nli-deberta-v3-large (0.52) failed to surpass the trivial baselines. As predicted by our EDA, the cross-attention mechanisms in these models learned to identify the high-lexical-overlap shortcuts present in the synthetic data, utterly failing to generalize to the abstract reasoning required by the human-annotated test set. Similarly, the 7-billion parameter Qwen-2.5-Instruct model, operating in a zero-shot capacity, hit a bottleneck at 0.56, highlighting the difficulty of extracting specific narrative alignment without targeted fine-tuning.

4.2 The Metric Learning Breakthrough

Transitioning to a metric-learning approach yielded immediate, substantial improvements. By fine-tuning the 22-million parameter all-MiniLM-L6-v2 utilizing Triplet Loss, we forced the model to construct global narrative representations rather than relying on pairwise lexical shortcuts. This single architectural shift raised Track A accuracy to 0.662 and Track B to 0.592.

4.3 Scaling with MNRL

To push our representations to the State-of-the-Art, we scaled our base architecture to BAAI/bge-large-en-v1.5 (330M parameters) and transitioned to Multiple Negatives Ranking Loss (MNRL). Utilizing a high-memory GPU cluster, we scaled our batch size, allowing the model to compute dozens of in-batch hard negatives simultaneously.

This configuration produced our champion model, achieving 0.680 accuracy on Track A and 0.652 accuracy on Track B. This result definitively proves the “Interaction vs. Representation Gap” for this dataset: models that are forced to compress narratives into robust, independent representations (Bi-Encoders) significantly outperform models that attempt to find token-level interactions (Cross-Encoders) when the underlying training data contains generative bias.

References

- [1] Sebastien Kawada and Dylan Holyoak. Cascademind at semeval-2026 task 4: A hybrid neuro-symbolic cascade for narrative similarity. *arXiv preprint arXiv:2601.19931*, 2026.
- [2] Max Upravitelev, Veronika Solopova, Jing Yang, Charlott Jakob, Premtim Sahitaj, Ariana Sahitaj, and Vera Schmitt. Multiperspectivity as a resource for narrative similarity prediction. *arXiv preprint arXiv:2603.22103*, 2026.